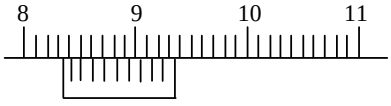
PHYSICS

SECTION – I
Single Correct Choice Type

This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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1. The length of a plate is measured with the help of a vernier calliper having zero correction – 0.02 cm. The observation is shown in figure. Then length of the plate from these observations is



(a) 8.30 cm (b) 8.34 cm (c) 8.32 cm (d) 8.36 cm

2. A particle is moving on x-axis according to the equation $x = \sin \omega t$. Consider the motion of the particle from $t = 0$ sec . to $t = 2\pi$ sec. Choose the correct statement.

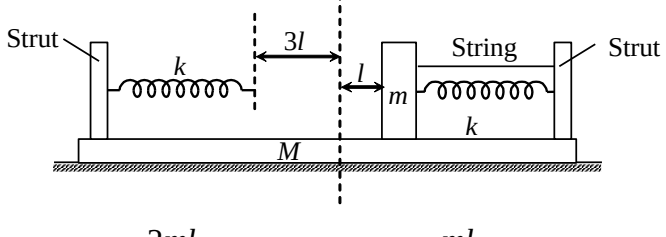
(a) Particle is retarding from $t = 0$ sec. to $t = \pi$ sec.

(b) Particle is retarding from $t = 0$ sec. to $t = \frac{\pi}{2}$ sec. and from $t = \pi$ sec. to $t = \frac{3\pi}{2}$ sec.

(c) Particle is retarding from $t = \frac{\pi}{2}$ sec. to $t = \frac{3\pi}{2}$ sec.

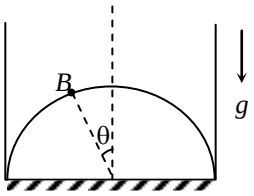
(d) Particle is retarding from $t = \frac{\pi}{2}$ sec. to $t = \pi$ sec. and from $t = \frac{3\pi}{2}$ sec. to $t = 2\pi$ sec.

3. A plank of mass M is placed on a smooth horizontal surface. Two light identical springs each of stiffness k are rigidly connected to struts at the ends of the plank as shown in the figure. When the springs are in their unextended position the distance between their free ends is $3l$. A block of mass m is placed on the plank and pressed against one of the springs so that it is compressed by l. To keep the block at rest it is connected to the strut by means of a light string. Initially the system is at rest. Now the string is burnt. Then the maximum displacement of the plank is



(a) $\frac{5ml}{m+M}$ (b) $\frac{2ml}{m+M}$ (c) $\frac{ml}{m+M}$ (d) $\frac{8ml}{m+M}$

4. A bead B of density ρ, which can slide on a friction less fixed semicircular structure of radius R is maintained at rest in an empty container as shown in figure. Now a liquid of density 2ρ is poured in the container upto height h ($h > R$) and the bead is released. Maximum contact force between bead and the structure at any time during the motion of the bead will be proportional to (θ is very small).



(a) $1 + \theta$ (b) $1 + \theta^2$ (c) $1 - \theta^2$ (d) $1 - \theta$

5. A monoatomic ideal gas undergoes a process $T^2 = 4P$. Molar specific heat of the process will be ($R \rightarrow$ universal gas constant)

(a) R (b) $\frac{5}{2}R$ (c) $\frac{R}{2}$ (d) $2R$

6. A block of mass M = 1 kg is dropped on a vertical spring, one end fixed to the ground. Height of the block from the free end of the spring is $\frac{5}{4}m$. Spring constant is $k = 5N/m$. After what time will the block come to its initial position again.

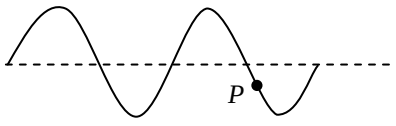
(a) $1+\frac{2\pi}{\sqrt{5}}+\frac{2}{\sqrt{5}}\sin^{-1}\left(\frac{2}{3}\right)$ (b) $1+\frac{2\pi}{\sqrt{5}}-\frac{2}{\sqrt{5}}\cos^{-1}\left(\frac{2}{3}\right)$

(c) $1+\frac{\pi}{\sqrt{5}}-\frac{2}{\sqrt{5}}\sin^{-1}\left(\frac{2}{3}\right)$ (d) $1+\frac{\pi}{\sqrt{5}}+\frac{2}{\sqrt{5}}\cos^{-1}\left(\frac{2}{3}\right)$

7. A point object is placed at the centre of curvature on the principal axis of a convex lens of focal length 10 cm. If a slab of some material of thickness 2 cm is introduced between the object and the lens magnification becomes half. Refractive index of material is (upto two significant digits)

(a) 0.16 (b) 0.17 (c) 0.18 (d) 0.19

8. A progressive wave heading in –ve *x* direction is maintained in a stretched string. At a certain instant snapshot of some part of string is taken which is as shown. Choose the correct direction of velocity of point *P* at this instant.



(a) upwards (b) downwards

(c) It may be upwards or downwards (d) Data insufficient

SECTION – II
Multiple Correct Choice Type

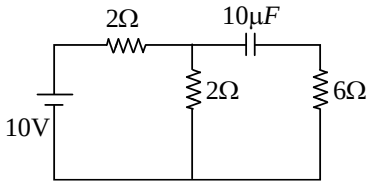
This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

---

9. Choose the correct option(s)

(a) Time constant of the circuit is 70 μsec

(b) In steady state, charge on the capacitor is 50 μC



(c) At any instant *t*, current through 6 Ω resistance is $\frac{5}{7}e^{-t/\tau}\ A\ (\tau = 70\,\mu\text{sec})$

(d) At any instant *t*, current through 6 Ω resistance is $\frac{3}{7}+\frac{5}{7}e^{-t/\tau}\ A\ (\tau = 70\,\mu\text{sec})$

10. A charge particle of specific charge $\left(\frac{q}{m}\right)$ 8 columb / kg is projected on a rough horizontal *X*-*Y* plane. Electric and magnetic fields are given by $\vec{E} = -5\hat{k}N/C$ and $\vec{B} = -10\hat{k}$ tesla respectively are present in the region. The particle enters in the region at (4, 0, 0) m with a velocity $100\,\hat{j}$ m/sec. If coefficient of friction between the particle and plane is $\mu = \frac{1}{3}$, then choose the correct option(s)

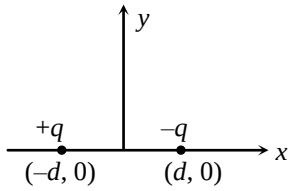
(a) The time after which particle comes at rest is 6 sec.

(b) Total work done by electric force on the particle is zero.

(c) Total distance covered by the particle is 300 m.

(d) Total work done by magnetic force on the particle is zero.

11. Two charges + q and – q are fixed on x-axis as shown. Consider a region in y-z plane $a^2 \le y^2 + z^2 \le b^2$. Choose the correct statement(s). ($a >>> d$)



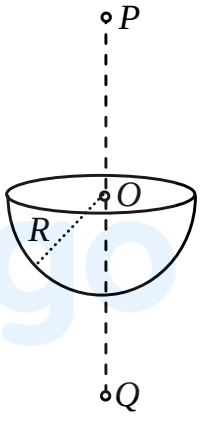
(a) Electric field anywhere in the given region is directed towards +ve x-axis.

(b) work done by the electric field in bringing a +ve test charge from $\left(0, \frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$ to $\left(0, \frac{a}{\sqrt{2}}, \frac{-a}{\sqrt{2}}\right)$ is zero.

(c) Electric potential throughout the given region is zero.

(d) flux crossing this surface is $\frac{qd}{\in_0}\left(\frac{1}{a} - \frac{1}{b}\right)$

12. Consider a hemispherical body of uniform mass density ρ and radius R as shown. P and Q are two points such that $OP = OQ = 2R$ as shown. Choose the correct statement(s)



(a) Magnitude of gravitational field intensity at P and Q are same.

(b) Gravitational field intensity at P and Q are unlike parallel.

(c) Magnitude of gravitational field intensity at Q is $\frac{\pi G \rho R}{6}$

(d) If E_0 is the magnitude of gravitational field intensity at Q then at P magnitude of gravitational field intensity is $\frac{\pi G \rho R}{3} - E_0$.

SECTION–III
Comprehension Type

This section contains 1 group of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

A closed cylinder insulated from the surroundings is divided into two equal parts by means of an adiabatic piston. The left part is filled with 32 g of O_2 and 4 g of He and right part is vacuum. The initial temperature of the mixture is T_0. One third of oxygen gas dissociates into atoms. The piston is released and the gas fills the whole volume of the cylinder. The piston is slowly displaced back to its original position. Neglect further dissociation of gaseous molecules during motion of the piston.

(Take $2^{0.56} \approx \sqrt{2}$, R = 8.3 J/K/mol, $\sqrt{2} = 1.41$).

13. The adiabatic exponent for the mixture is

(a) 1.56 (b) 1.5

(c) 1.8 (d) 1.53

14. The temperature of the gas when piston returns back to its initial position is

(a) $T_0\sqrt{2}$ (b) $\frac{T_0}{\sqrt{2}}$

(c) $2T_0$ (d) $\frac{T_0}{2}$

15. The increase in internal energy of the gas is (in J)

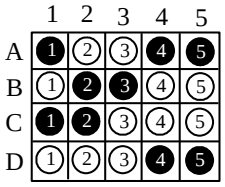
(a) 14.18 T_0 (b) 14 T_0

(c) 10 T_0 (d) zero

SECTION–IV
Matrix–Match Type

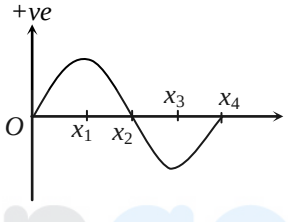
This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



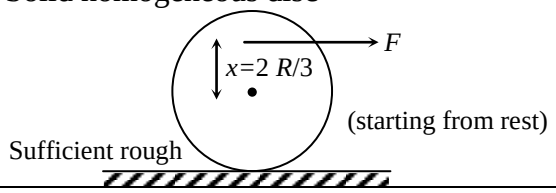
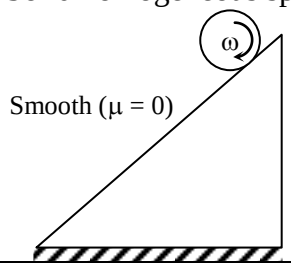
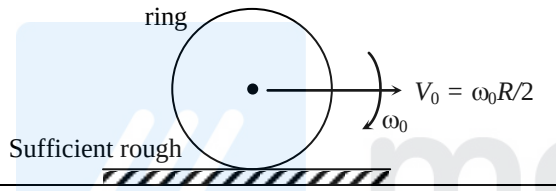
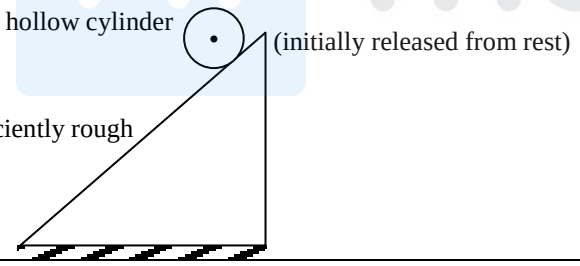
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1. The figure represent a sinusoidal wave whose general equation is $y = aA\sin(kx + b\omega t)$ where A, k, ω, are some arbitrary constants. P is a general point on the wave at a horizontal distance x from origin O, and x_1, x_2, x_3, x_4 are some special point on x-axis dividing wavelength in four equal intervals. V_p is the particle velocity of point P. Then match the following options ($|a| = |b| = 1$)



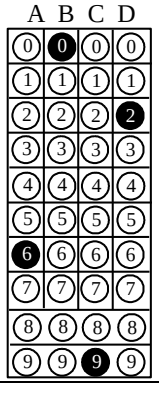
Column I		Column II	
(A)	V_P is +ve	**1.**	if $a > 0, b < 0, 0 < x < x_1$
(B)	V_P is –ve	**2.**	if $a < 0, b < 0, x_1 < x < x_2$
(C)	Wave is travelling along +ve x axis	**3.**	if $a > 0, b > 0, x_2 < x < x_3$
(D)	Wave is travelling along –ve x-axis	**4.**	if $a < 0, b > 0, x_2 < x < x_3$
		5.	if $a > 0, b < 0, x_1 < x < x_2$

2. Column – I shows few systems. In column – II few statements are given about those systems. Choose proper statement(s) for each system.

Column I	**Column II**
(A) Solid homogeneous disc 	**1.** friction acting along the direction of motion of centre of mass.
(B) Solid homogeneous sphere 	**2.** friction acting opposite to the direction of motion of centre of mass.
(C) 	**3.** friction acting is zero.
(D) 	**4.** Net work done by friction is zero.
	5. Net work done by friction is negative.

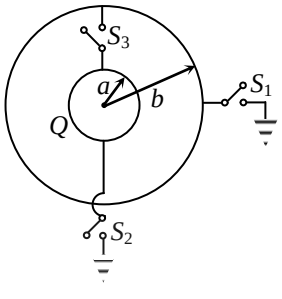
SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



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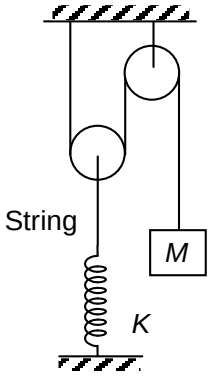
1. The figure shows a conducting sphere 'A' of radius 'a' which is surrounded by a neutral conducting spherical shell B of radius 'b' ($>a$). Initially switches S_1, S_2 and S_3 are open and sphere 'A' carries a charge Q. First the switch 'S_1' is closed to connect the shell B with the ground and then opened. Now the switch 'S_2' is closed so that the sphere 'A' is grounded and then S_2 is opened. Finally, the switch 'S_3' is closed to connect the spheres together. The heat (in Joule) which is produced after closing the switch S_3 is $0.2x$. Find x [Consider b = 4 cm, a = 2 cm and Q = 8 μC]



Space for work

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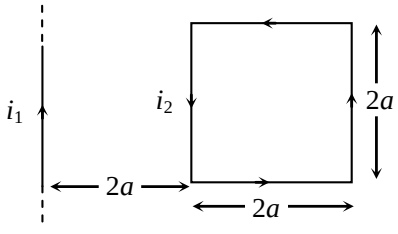
2. Consider the given system. All strings, pulleys and spring are ideal, mass M is in equilibrium. Maximum amplitude of oscillation of block in vertical direction is $\alpha\frac{Mg}{k}$. Find value of α.



Space for work

mathongo

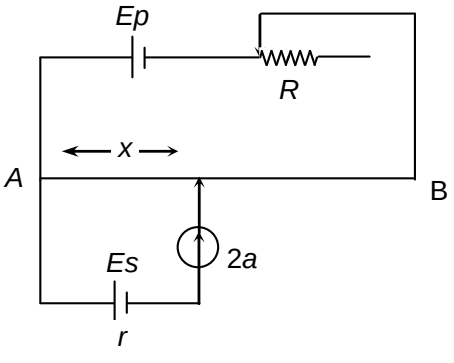
3. An infinitely long straight wire carries a current $i_1 = 1$ amp. A rectangular current loop placed in the same plane of shown dimensions carries a current $i_2 = 1$ amp anticlockwise. Magnitude of net force on the loop is found to be $\beta\frac{\mu_0}{4\pi}$. Find value of β.



Space for work

mathongo

4. Consider a potentiometer circuit. Primary cell is ideal. The length of potentiometer wire is 1m and the resistance per unit length of potentiometer wire varies with length as $\lambda = 2x\,\Omega/m$. Where x is distance from end A. Resistance of Rheostat varies with time as $R = t^2\Omega$. Null deflection point for secondary cell is obtained at $x = \frac{1}{2}$ m and at $t = 1$ sec. If emf of secondary cell is $\frac{1}{\gamma}$ times of emf of primary cell. Find γ.



Space for work

mathongo